CLASSIFICATION OF ELEMENTS AND PERIODICITY IN THESE PROPERTIES.

Dobereiner's law of Triads

→ Dobereiner formed group of 3 elements (called triads) having similar properties where the atomic wt. of middle element was an avg. of 1st & 3rd element.

→ (Based on atomic wt.)

Eg:- ^{7}Li ^{23}Na ^{39}K

^{40}Ca ^{88}Sr ^{137}Ba

$^{35.5}Cl$ ^{80}Br ^{127}I

Drawback: Many elements could not fit into the triad classification.

Newland's law of octaves

→ Like every 8th note repeats itself, Newland arranged elements in increasing order of their at. wt and suggested that every 8th element repeats its properties.

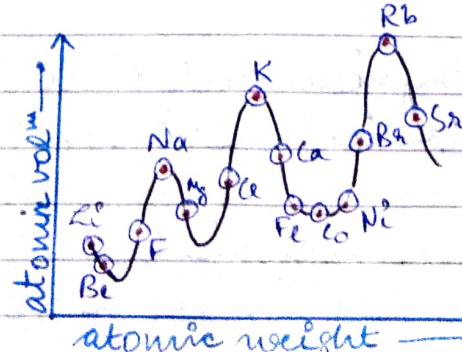
→ (Based on atomic wt)

Drawback: Didn't think about elements which were not discovered during that time.

After, the discovery of noble gases, it was found out that every 9th element had similar properties rather than every 8th.

Lother - Meyer's curve (Based on atomic wt.)

→ Elements of similar properties occupy similar positions in a curve plotted by him as atomic volm v/s atomic wt.



Features: ① Peaks (or crests) were occupied by alkali metals

② Troughs (or lowest pts.) are occupied by d-block metals

③ Alkaline earth metals occupy nearly the mid points of descending portion.

④ Halogens occupy the mid pts. of ascending portions of the curve.

Drawback: Lack of practical utility & difficult to remember.

Mendeleev's periodic table

→ First organized classification in tabular form
→ (Based on atomic wt)
→ Periodic table → 7 horizontal rows (periods)
→ 9 vertical columns (groups).
→ He arranged elements in increasing order of atomic wt. and showed that elements of similar properties appear in the same group.

Merits: ① First classification in organized tabular form
② Left vacant spaces in his periodic table for the elements that were not discovered during that time. He named these elements as eka-aluminium (gallium) and eka-silicon (germanium)

Drawbacks: ① No clear position of hydrogen.
② Dissimilar elements were placed in same groups
Eg:- Cu, Ag, Au (coinage metals) were placed along with alkali metals.
③ Manganese was placed along with halogens.

Mendeleev's periodic law:- properties of elements are the periodic functions of their atomic wt.

Moseley's Periodic law: Physical and chemical properties of elements are the periodic function of their atomic numbers.

→ Moseley bombarded highly energetic beam of e^- on a metal and studied the frequency of x-rays emitted

* 94 naturally occurring elements. (neptunium & plutonium like actinium & protactinium are also found in pitch blende — an ore of uranium)

from metal. He found that frequency of emitted x-rays are closely related with atomic no. of metal (& not with mass)

$$\sqrt{\nu} = a(Z - \sigma)$$ where, ν = frequency of emitted x-rays.

Z = atomic no. a = proportionality constant.
σ = screening constant or shielding constant.

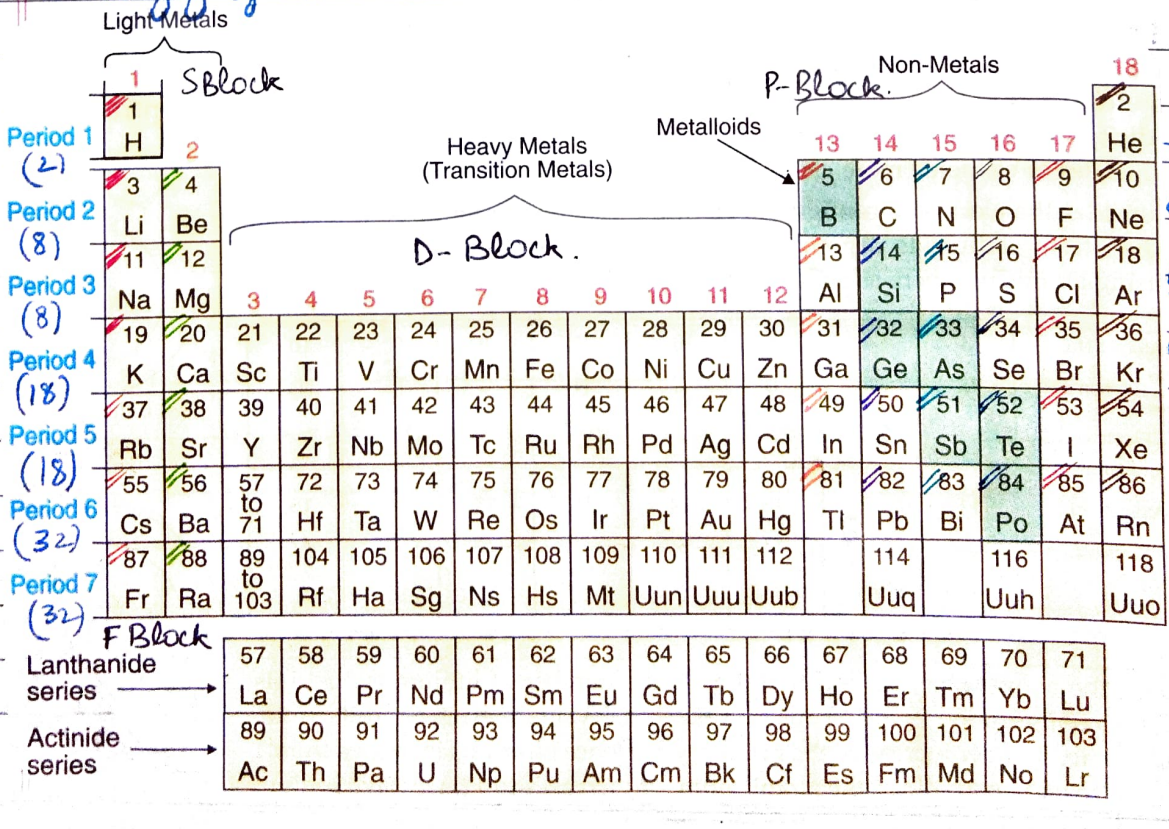
→ Atomic no. (Z) is the more fundamental property which decides physical & chemical properties of elements

Modern form of periodic table

→ Developed by Bohr, Bury, Rang and Werner based on Moseley's periodic law.

→ Consists of → 7 horizontal rows called periods.
→ 18 vertical columns called groups.

→ Period no. represents max^m principal quantum no. in e^- config. of element.



→ outer electronic config.

s block: (ns^{1-2}) d-block: $(n-1)d^{1-10}\ ns^{0-2}$

p block: (ns^2np^{1-6}) f-block: $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$.

→ nearly 78% of elements are metals, remaining are mostly non-metals & metalloids.

→ Metalloids :- They have properties of metals as well as non-metals.

→ No. of elements in a period is decided by possible orbitals as following (based on Aufbau's rule).

ns, np, (n-1) d, (n-2) f, (n-3) g, (n-4) h.

Eg:- 4th period ⇒ n=4 | 4s → 1, 4p → 3, 3d → 5 → 9 orbitals × 2 ⇒ 18 elements.

6th period ⇒ n=6 | 6s → 1, 6p → 3, 5d → 5, 4f → 7 → 16 orbitals × 2 ⇒ 32 elements.

NOTE:- ① Periodic table can be divided into 3 parts →

- Representative elements → s & p-block elements (Grp No: 1, 2, 13-18)
- Transition elements → d-block elements (all metals) * Zn, Cd & Hg (Grp. No=12) are d-block elements but are not regarded as transition elements.
- Inner transition elements (f-block elements, all metals)

② Typical elements:- 3rd pd elements except noble gas. Other elements in same grp. have similar properties to others in the group.

(Typical elements)

Na Mg Al Si P S Cl

→ 2nd pd. elements (Li, Be, B, C, N, O & F) are not chosen as typical elements because they show different properties (exceptional behaviour) from rest of the elements of the same group.

It is due to →

(i) small size
(ii) high electronegativity
(iii) absence of d-orbitals in valence shell in 2nd period of elements.

International union of pure and applied chemistry.

IUPAC nomenclature of elements having Z > 100 :-

0 - nil	n	Temporary names given to new elements
1 - un	u	until their names are finalised.
2 - bi	b	→ A suffix 'ium' is added.
3 - tri	t	Z = 105 = unnilpentium.
4 - quad	q	Z = 117 = ununseptium.
5 - pent	p	
6 - hex	h	
7 - sept	s	
8 - oct	o	
9 - enn	e	

101 - Mendelevium (Md)
102 - Nobelium (No)
103 - Lawrencium (Lr)
104 - Rutherfordium (Rf)
105 - Dubnium (Db)
106 - Seaborgium (Sg)
107 - Bohrium (Bh)
108 - Hassium (Hs)
109 - Meitnerium (Mt)
110 - Darmstadtium (Ds)
111 - Roentgenium (Rg)
112 - Copernicium (Cn)
113 - Nihonium (Nh)
114 - Flerovium (Fl)
115 - Moscovium (Mc)
116 - Livermorium (Lv)
117 - Tennessine (Ts)
118 - Oganesson (Og)

104 → Americans named it Rutherfordium & soviets named it Kurchatovium.

Predicting Group No., period no., block of an element

→ Block of an element is the orbital which recieves last electron (exception : He)

→ Period no is the principal quantum no. of outermost shell.

→ IUPAC grp. No.

For s-block → Grp. no. = no. of ns e^-.

For p-block → Grp. no. = no. of np e^- + ns e^- + 10.

For d-block → Grp. no. = no. of (n-1)d e^- + no. of ns e^-

For f-block → Grp no. = 3 (fixed).

→ Effective nuclear charge (Z_{eff}):
(screening effect or shielding effect)

→ In multi electron species, intervening e^- (inner e^-s or e^- b/w nucleus & valence shell) reduce the force of attraction of nucleus on outer e^-. This reduction in force of attraction by inner e^- is called screening effect.

→ Inner e^- shield outer electrons from the attractive influence of nucleus.

$$Z_{eff} = Z - \sigma$$ where, Z = atomic no.

σ = sheilding constant / screening constant.

For single e^- species →

$$E_n = -13.6 \times \frac{Z^2}{n^2} \text{ eV}.$$

considering shielding effect → ⇒ $\sigma \uparrow$, $Z_{eff} \downarrow$.

For multi e^- species,

$$E_n = -13.6 \times \frac{Z_{eff}^2}{n^2} \text{ eV}.$$

Factors affecting sheilding constant →

① Shielding is mainly caused by same shell, inner e^- Outer e^- are not effective in causing shielding.

② Closer is the e^- to the nucleus, greater will be the

shielding caused by it. (This is due to penetration power)
For same shell (n),

$ns > np > nd > nf$

decreasing shielding caused (per e^- basis)

→ Calculation of shielding constant by Slater's rule.

(a) For ns, np e^-

(i) Write config. as →

(1s)(2s,2p)(3s,3p)(3d),(4s,4p)(4d)(4f), - - -

(ii) Groups right to (ns,np) do not contribute in σ

(iii) All the other e^- contribute 0.35 towards σ.
(However, note that in case of 1s, contribution is 0.3)

(iv) All the (n−1) shell e^- contribute 0.85 towards sigma

(v) Rest all inner e^- contribute 1 towards σ.

Eg:- Na: $(1s^2)(2s^2 2p^6)(3s^1)$.

σ on 3s e^- = $8 \times 0.85 + 2 \times 1$.

⇒ 6.80 + 2 = 8.8.

Z_{eff} on 3s = 11 − 8.8 = 2.2.

σ on 2p e^- = $7 \times 0.35 + 2 \times 0.85$

⇒ 2.45 + 1.70 = 4.15

Z_{eff} on 2p = 11 − 4.15 = 6.85.

(b) For nd or nf e^-

(i) Write config. as above.

(ii) e^- to the right of nd or nf do not contribute in σ.

(iii) All the other e^- of nd or nf contribute 0.35 towards σ.

(iv) All inner e^- contribute 1 towards σ.

Eg. - $_{30}Zn$: $(1s^2)(2s^2 2p^6)(3s^2 3p^6)(3d^{10})(4s^2)$

σ on 3d $e^- = 9 \times 0.35 + 18 \times 1$

$\Rightarrow 3.15 + 18 = 21.5$.

$Z_{eff} = 30 - 21.15 = 8.85$.

NOTE:- $_2He$: $1s^2$

$\sigma = 1 \times 0.3 = 0.3$

$Z_{eff} = 2 - 0.3 = 1.7$.

Periodicity in properties (Trends)

① Atomic radius (or atomic size).

↳ It is the distance b/w nucleus & outermost e^- in an atom.

↳ Exact size of atom can not be determined as →

(i) a single atom can't be isolated.

(ii) position of e^- can't be exactly determined (Heisenberg's uncertainity principle)

↳ Indirectly, size (radius) of an atom of an element is determined indirectly using bond data (by X-ray diffraction or spectroscopy).

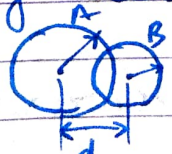
[A] Covalent radius (r_{cov}):

It is One half of internuclear distance b/w atoms in a homonuclear diatomic molecule (O_2, N_2, H_2, Cl_2 etc)

$$r_{cov} = \frac{d}{2}.$$

However, covalent radius is slightly smaller than the actual radius.

In case of heteronuclear molecule,



$$d_{AB} = r_A + r_B - 0.09|\chi_A - \chi_B|$$

↳ Stevenson's formula

d_{AB}, r_A, r_B → in Å.

χ_A, χ_B → electronegativity in pauling's scale.

↳ Covalent radius mainly applies to now metals as they form covalent bonds.

↳ B/w same pair of atoms.

$\equiv < = < -$ (Increasing order of covalent radii).

↳ Single bond covalent radii is most accurate among these type of radii b/w a pair of atoms.

[B] Metallic radius (r_{met})

It is one half of internuclear distance b/w two metals atoms in a metal lattice.

↳ Mainly applies to metals.

↳ Metallic radius is slightly larger than the actual size (radius of metal atom).

[C] Van der waal's radii (r_{van})

↳ Also called as non-bonding radius

↳ Mainly used for noble gases like He, Ne, Ar etc which neither form a covalent bond nor exist as metallic lattice.

↳ It is one half of the internuclear distance of atoms in closest proximity in solid or liquid state.

$(r_{van})_{Ne} = \frac{d}{2}$.

→ Van der Waal's radius is much larger than the actual radius (highly erroneous).

NOTE:- For a specific element.

$r_{van} >> r_{met} > r_{cov}$

↳ Factors affecting atomic radius:

① Z_{eff} → as Z_{eff} increases, atomic size decreases.

② No. of shells in atom (or principal quantum no.) (n) as n increases, size also increases.

Trends of Z_{eff} (in s & p block)

→ ongoing from left to right, Z_{eff} increases

→ & down the group, Z_{eff} remains nearly constant

Trends of atomic radius (in s & p block)

→ along the period, atomic size ↓es as Z_{eff} increases

→ down the group, atomic size ↑es as no. of shells increase.

Exceptions

① In 2nd, 3rd, 4th pd. size of noble gas is found to be largest in the pd. It is due to van der waal's radius used for noble gas.

→ on uniform scale (i.e all radii are van der waal's radii), there will not be any exception in radii of elements of a pd.

② Atomic radius of Ga < Al. It is due to poor shielding caused by 'd' electrons that are filled in Ga.

$$ns > np > nd > nf$$

decreasing order of shielding caused.

→ 'd' & 'f' e⁻ are poor shielding e⁻

Al: $1s^2 2s^2 2p^6 3s^2 3p^1$

Ga: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^1$

↳ Z_{eff} is high & hence radius decreases.

③ Lanthanoid Contraction.

3d	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn.	
4d	Y	Zr	Nb	Mb	Tc	Ru	Rh	Pd	Ag	Cd	
5d	La(*)	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	

↑ Lanthanide contraction is observed here.

No lanthanoid contraction → Sc < Y < La (size →)

Lanthanoid contraction → Ti < Zr ≅ Hf ; Cu < Ag ≅ Au

↳ When we move from 3d to 4d series (vertically down), atomic size increases as expected)

↳ When we move from 4d to 5d series (vertically down), atomic size after La do not increase, rather slightly decrease or remains nearly same This effect is seen after lanthanoid series & hence called as lanthanoid contraction.

↳ Reason for lanthanoid contraction is the poor shielding of 14 e^- filled in f subshell just before Hf (Hafnium). So, Z_{eff} increases very sharply from La to Hf.

↳ Effect of lanthanoid contraction is also applicable in ionization enthalpy & further into p-block.

Trends of atomic radii (in 3d series)

↳ Size decreases upto Fe, then the size of Fe, Co & Ni almost becomes same and then increases till Zn.

* Atomic radii of Cu & Zn are slightly larger due to greater repulsion among e^- in fully filled d-subshell.

Ionic radius

↳ Size of cation is smaller than its neutral atom.

↳ Size of anion is larger than size of its neutral atom

* $F \longrightarrow F^-$: q pm → ≅ 2q pm.

* $Na \longrightarrow Na^+$: x pm → $\frac{4x}{5}$ pm.

$Mg > Mg^+ >> Mg^{2+}$ ← one complete shell is removed.

Smallest atom: H — Smallest cation = H^+

Largest atom: Cs — Smallest anion = F^-

* $r_{H^-} \cong r_{Br^-}$. This means that H^- is enough large anion.

Eg:-

$F \longrightarrow F^-$

$\frac{e}{p} = \frac{9}{9} \longrightarrow \frac{e^-}{p} = \frac{10}{9}$

% increase $= \frac{10/9 - 9/9}{1} \times 100$

$= 11\%$.

$H \longrightarrow H^-$

$\frac{e}{p} = \frac{1}{1} \longrightarrow \frac{e}{p} = \frac{2}{1}$

% increase $= \frac{2-1}{1} \times 100$

$= 100\%$

* H^- is an unstable anion, therefore saline hydrides (or ionic hydrides) are powerful reducing agents. (eg. LiH, NaH, CaH_2, $LiAlH_4$, etc)

Isoelectronic species :- Ions having same no. of total e^-.

	N^{3-}	O^{2-}	F^-	Na^+	Mg^{2+}	Al^{3+}
$\frac{Z}{e} =$	7/10	8/10	9/10	11/10	12/10	13/10.

→ higher is $\frac{Z}{e}$ ratio, small is the size (radius) among isoelectronic species.

$N^{3-} > O^{2-} > F^- > Na^+ > Mg^{2+} > Al^{3+}$

decreasing order of ionic radii.

Remember

$r_{Li^+} > r_{Mg^{2+}} > r_{Al^{3+}}$

0.76Å 0.72Å 0.53Å.

→ decreasing size.

Q Compare size of O^{2-}, S^{2-}, Cl^-, F^-.

$S^{2-} > Cl^-$

$O^{2-} > F^-$

$S^{2-} > Cl^- > O^{2-} > F^-$

→ decreasing size

② Ionization energy (IE) (defined at 0K).

→ Energy req. to remove the most loosely bound e^- from a gaseous atom to form gaseous cation.

→ Usually most loosely bound e^- is the outermost e^- which experiences least Z_{eff}.

$$M_{(g)} \longrightarrow M^+_{(g)}, \text{ energy req.} = (IE)_1[M_{(g)}]$$

$$M^+_{(g)} \longrightarrow M^{2+}_{(g)} + e^-, (IE)_2[M_{(g)}], (IE)_1[M^+_{(g)}].$$

NOTE:- For a given element, successive IE goes on increasing

For element M,

$$IE_1 < IE_2 < IE_3 - \text{---}$$

→ In thermodynamic convention, energy involved in a process or reaction are termed as enthalpy change (ΔH).

→ Endothermic rxn (ΔH = +ve) (energy is absorbed)

→ Exothermic rxn (ΔH = -ve) (energy is evolved)

→ (defined at any temp^r)

Ionization enthalpy → (ΔH_{IE}): It is the enthalpy change when 1 mole of gaseous atoms of an element are converted into gaseous atom.

$$M_{(g)} \longrightarrow M^+_{(g)} + e^- \quad (\Delta H_{IE})_1[M_{(g)}]$$

(1 mol)

Relationship/ve IE & ΔH_{IE}.

$$M_{(g)} \longrightarrow M^+_{(g)} + e^-$$

Taking M, M^+ & e^- to be monoatomic.

Using Kirchoff's eqn →

$$\frac{\Delta H_{T_2} - \Delta H_{T_1}}{T_2 - T_1} = (\Delta C_p)_r$$

$$\frac{\Delta H_{IE} - IE}{T-0} = \frac{5}{2}R + \frac{5}{2}R - \frac{5}{2}R.$$

$$\frac{\Delta H_{IE} - IE}{T-0} = \frac{5}{2}R.$$

$$\boxed{\Delta H_{IE} = IE + \frac{5}{2}RT}$$

ΔH_{IE} : J/atom ; IE : J/atom ; $\frac{5}{2}RT$: J/atom.

Factors affecting I.E ⟶

① as size increases, I.E decreases.

② Greater the value of Z_{eff}, greater is the IE.

③ special electronic config.
→ Half filled and fully filled config. are highly stable and therefore more energy is needed to remove e^- from such config.

④ Penetration power.
For same shell (n),
$ns > np > nd > nf$
→ decreasing penetration power.

Eg:- $2s > 2p$
↑ removal of e^- from 2s requires more energy than that from 2p because penetration power of 2s is more.

Trends of IE in periodic table
→ From left to right, ionization energy increases.
→ down the group, ionization energy decreases.

Exceptions

$IE_{Be} > IE_B$

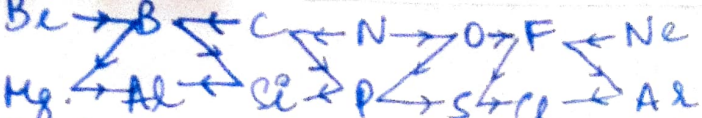
Li	[Be]	B	C	[N]	O	F	Ne
Na	[Mg]	Al	Si	[P]	S	Cl	Ar.

① higher amt. of energy is req. to remove e^-.

$IE_{Mg} > IE_{Al}$

$IE_p > IE_s$

$IE_N > IE_O$ (half filled stable config.)



Ne > F > Ar > Cl → decreasing IE.

② IE of Ga is slightly greater than the IE of Al.
→ due to poor shielding of 10 e^- in d-subshell before Ga.

③ Lanthanoid contraction. → due to poor shielding caused by 14 e^- filled in 4f subshell just after La.

After La, $IE_{5d} > IE_{4d}$
$IE_{5d} > IE_{3d}$

→ The effect of lanthanoid contraction extends upto Pb (in p-block) in ionization energy.

$IE_{Pb} > IE_{Sn}$.

Applications of IE →

① Elements with low IE are electropositive in nature (said to have high metallic character)
→ Along the period, electropositive nature or metallic nature decreases.
→ Down the group, electropositive or metallic nature increases.

$Li < Na < K < Rb < Cs$
→ increasing metallic character.

② Elements which have low IE, form basic oxides.
In general, alkali metal oxides are more basic than the oxides of alkaline earth metal oxides.
→ Along the pd, basic nature of oxides decreases.
→ Down the grp, basic nature of oxides increases.

③ Since metals easily lose e^- (i.e undergo oxidation), hence metals act as powerful reducing agent.

$M_{(s)} \longrightarrow M^+_{(aq)} + e^-$

→ On the basis of IE, Cs is expected to be more powerful reducing agent, However, the fact is that (Li) is most powerful reducing metal (or element)

→ High reducing power of Li is due to very high hydration

energy (released) of Li^+ (small sized cation).

(2) Difference in successive IE helps in predicting stable oxd^n state of an element

→ In general, a high energy diff. in successive IE indicate, greater stability of the lower oxd^n state.

→ A low diff. in successive IE indicate greater stability of higher oxd^n state.

(3) Electron affinity (EA) → It is the energy released when an e^- is added to an isolated gaseous atom to form gaseous anion.

$$X_{(g)} + e^- \longrightarrow X^-_{(g)}, \text{ energy released} = (EA_1)_X.$$

* In most elements, EA is +ve, i,e energy is released when e^- is added.

However, in few cases, EA is -ve, when energy is absorbed on adding e^-.

Electron gain enthalpy (ΔH_{eg}): It is the enthalpy change when 1 mol of gaseous atoms of an element is converted into gaseous anion.

$$\underset{1\,mol}{X_{(g)}} + e^- \longrightarrow X^-_{(g)} \quad \Delta H_{eg_1}$$

ΔH_{eg_1} is -ve for most elements as energy is released on adding an e^-.

However, in some elements ΔH_{eg} (or ΔH_{eg_1}) is +ve in few cases where energy is absorbed.

→ In general, (ΔH_{eg}) is the negative of electron affinity in same energy units.

$$X_{(g)} + e^- \longrightarrow X^-_{(g)}$$

$$\boxed{\Delta H_{eg} = -EA - \frac{5}{2}RT}$$

EA → defined at '0' K.

ΔH_{eg} → defined at any temp^r.

→ Note, that in comparison purpose or for comparing trends in periodic table, we use both the terms interchangeably.

NOTE:- $O_{(g)} + e^- \longrightarrow O^-_{(g)}$, $EA_1(O_{(g)}) = +ve$, energy is released.

$O^-_{(g)} + e^- \longrightarrow O^{2-}_{(g)}$, $EA_2(O_{(g)}) = -ve$, energy is absorbed heavily.

↳ $O_{(g)} + 2e^- \longrightarrow O^{2-}$, $EA_1 + EA_2 = -ve$, always.
not only for oxygen, but general also for monovalent & bivalent anion.
Or, $\Delta H_{eg_1} + \Delta H_{eg_2} = +ve$ (always).

→ PE of attraction is -ve.
→ PE of repulsion is +ve.
If nucleus attraction for newly added e^- is dominant over repulsion by existing e^-, then energy is released
i,e $EA = +ve$, $\Delta H_{eg} = -ve$ ζ.

Factors affecting EA:-
① Size increases, electron affinity decreases.
② Z_{eff} increases, electron affinity increases.
③ Half filled & fully filled configurations are highly stable and have little tendency to add e^-. In other words, EA is mostly -ve (i,e energy is absorbed) in such configurations.

General trends of EA in periodic table.
→ Along the period, E affinity increases.
→ Down the group, EA decreases.
→ Trends of EA are less regular than that of ionization energy.

Exceptions Of EA:-

① -ve EA is for the following notable elements →

Be N He Ne Ar Kr Xe — -ve EA or +ve ΔH_{eg}.

Mg / ↳ due to highly stable fully filled config.

due to half filled stable config -

→ only 'N' has -ve EA in pnictogens or +ve ΔH_{eg}.

② EA of 3rd period 'p-block' elements are greater than the corresponding 2nd period p-block elements.

B C N O F → 2nd pd.

Al Si P S Cl. → 3rd pd.

Cl > F, S > O, P > N, Si > C, Al > B.

* Highest EA → chlorine.

→ It is due to exceptionally small size of 2nd pd. p-block elements. When e^- is added, a very heavy repulsion is created by existing valence e^- of 2nd pd. p-block elements due to high e^- density (again due to very small size).

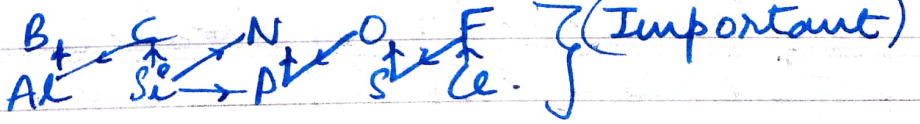
So, lesser energy is released than expected for 2nd pd 'p-block' elements.

③ 'O' in chalcogens has smallest EA value.

S > Se > Te > Po > O → decreasing EA values.

Cl > F > Br > I → decreasing EA values.

⇒ For EA comparisons,



(Important)

N < P < O < S < F < Cl → increasing EA.

B < Al < C < Si → increasing EA.

→ The smallest value of IE of an element (Cesium) is even greater than the highest value of EA of an element (chlorine)

Applications of EA →

① It is used in the calculation of electronegativity of an element (using mulliken's scale).

② Generally, non-metals (halogens, chalcogens) have high EA values.

→ High EA of an element represents high tendency to gain e^- and therefore such elements act as good oxidizing agents.

→ On this basis, Cl_2 is expected to be the most powerful oxidizing element in periodic table, however, F_2 is most powerful oxidizing element in periodic table.

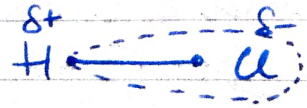
→ Highest oxidizing power of F_2 is due to mainly, high hydration energy of small F^- ion and low bond dissociation energy of F_2.

$Cl_2 > Br_2 > F_2 > I_2$ decreasing bond energy order →

④ Electronegativity (χ) → It is defined as the ability of an atom to pull shared pair of e^- towards itself in a molecule.

→ Electronegativity is not the property of an isolated atom, rather it is the property of an atom in a covalent bond, it is not a measurable quantity.

* Property of an isolated atom are ionization energy & electron affinity.



→ polar covalent bond.
(covalent molecule).
(polar).

→ It is determined in various scales.

① Pauling scale of EN

$A^{\delta+} - B^{\delta-}$ ← more EN polar covalent bond.

$$\chi_A - \chi_B = 0.208\sqrt{BE_{A-B} - \sqrt{BE_{A-A} \times BE_{B-B}}}$$

BE_{A-B}, BE_{A-A} and BE_{B-B} are in kcal/mol.

χ_A χ_B → Pauling electronegativities of A & B.

→ Originally pauling assigned EN of H-atom to be 2.1 & determined EN of other elements. (However, NCERT mentions that reference chosen by pauling was fluorine as 4.0)

Pauling values of EN,

F > O > N ≅ Cl > Br > C ≅ S ≅ I > H ≅ P > B > Li > Cs.

4.0 3.5 3.0 3.0 2.8 2.5 2.5 2.5 2.1 2.1 2.0 1.0 0.7

② Mulliken scale of EN.

Acc. to mulliken, EN of an element is the avg. of IE and EA in eV. $EN_{mulliken} = \frac{IE + EA}{2}$.

$$EN_{pauling} = \frac{EN_{mulliken}}{2.8}$$

NOTE:- Electronegativity of an element may be different in different molecules.

Factors affecting EN →

① Greater the size, lower is the EN.

② Greater the Z_{eff}, greater the EN

③ Ond^{n} no: → Higher the +ve ond^{n} state, higher will be the EN. Eg:- $Fe < Fe^{2+} < Fe^{3+}$.

→ Higher the -ve oxid^{n} state, lower will be the EN.

Eg:- $O > O^{2-}$

$F > F^{-}$

④ State of hybridization:

Greater the % s-character in the hybridized atom, greater is the EN of an atom.

(This is due to greater penetration power of $ns > np$)

Eg:-

	C_2H_6	C_2H_4	C_2H_2
	(sp^3)	(sp^2)	(sp)
%s =	25%	33.3%	50%

$C_2H_2 < C_2H_4 < C_2H_2$

→ increasing EN of 'C' atom

General trends of EN →

→ Along the period, EN increases.

→ Down the group, EN decreases.

Eg:- $F > Cl > Br > I > At$.

$O > S > Se > Te > Po$.

$Li > Na > K \geq Rb > Cs$.

→ decreasing EN.

Applications of EN →

① % ionic character in a polar covalent bond:-

Eg:-

$A^{\delta+} - B^{\delta-}$

↑ greater EN

polar covalent bond.

Δ(EN) ↑, greater will be the % ionic character.

Acc. to Hanny & Smith, % ionic character in a bond A-B can be calculated by →

$= 16|\chi_A - \chi_B| + 3.5(\chi_A - \chi_B)^2$.

Acc. to Hanny & Smith, if $\Delta(EN) = 2.1$, % ionic character is 50%.

→ If $\Delta(EN) > 2.1$, bond is said to be predominantly ionic bond.

→ If $\Delta(EN)$ is < 2.1, bond is predominantly covalent.

More appropriately, acc. to Pauling →

If $\Delta(EN) = 1.7$, then 50% ionic character exists.

If $\Delta(EN) > 1.7$, bond is predominantly ionic.

If $\Delta(EN) < 1.7$, bond is predominantly covalent.

Eg:- HCl, $\Delta(EN) = 3.0 - 2.1 = 0.9 < 1.7$
(polar covalent bond).

NaCl, $\Delta(EN) = 3.0 - 0.9 = 2.1 > 1.7$
(ionic bond).

② Higher is EN difference, in general, bond will be shorter & stronger and will be covalent.

Eg:- H-F > H-Cl > H-Br > H-I
→ decreasing bond energy.

H-F < H-Cl < H-Br < H-I
→ increasing bond length.

③ Determination of Ond^{n} no:-

Eg:- (i) $\overset{+1}{H}-\overset{-1}{Cl}$
Ond^{n} no. of H = +1
Ond^{n} no. of Cl = −1

(ii) $\overset{-2}{O}=\overset{+2}{C}=\overset{-2}{O}$
Ond^{n} no. of C = +4
Ond^{n} no. of O = −2.

④ Acidic or basic nature of XOH molecule:-

$X-O-H \xrightarrow{H_2O} X^{+}_{(aq)} + OH^{-}_{(aq)}$ (base).

(when $EN_X < EN_H$) Eg:- NaOH, KOH.

$X-O-H \xrightarrow{H_2O(l)} XO^-_{(aq)} + H^+_{(aq)}$ (acid).

(where, $EN_X > EN_H$)

Eg:- HClO, HBrO → (hypobromous acid)
↳ (hypochlorous acid).

→ More polar bond (in general), dissociates more easily in water due to very high dielectric constant of water ($\cong 79$).

⑤ Elements with high EN, have high EN character.
Eg:- 'F' is the most electronegative element.

→ Periodicity in chemical properties depend on outer electronic config. Elements with similar electronic config. (particularly outer electronic config) have similar chemical behaviour.

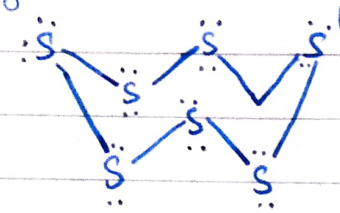
→ Periodicity in valence also depends on electronic config.
Valence → no. of outermost e^- / 8 − no. of outermost e^-)
for representative elements (1, 2, 13–18 grp).

Grp No:-	1	2	13	14	15	16	17	18
outermost e:-	1	2	3	4	5	6	7	8.
Valence :-	1	2	3	4	5, 3	6, 2	7, 1	0, 8.

→ Nowadays, a more appropriate term, called Oxid^n no. is more commonly used.

Valency
- → Covalency (in covalent compounds)
  ↳ No. of covalent bonds formed.
- → Electrovalency (in ionic compounds).
  ↳ No. of e^- lost or gained by an element.

Eg:- (i) $[Al(H_2O)_6]^{3+}$ covalency of Al = 6.
Oxd^n no. of Al = +3.

(ii) S_8 covalency = 2

Oxd^n no: - 0.

(iii) BaCl$_2$
Electrovalency of Ba = 2.
Electrovalency of Cl = 1.

Trends of acidic and basic nature of oxides.

→ Acidic oxides are anhydrides of oxyacids.

Eg:- $H_2SO_4 \xrightarrow{-H_2O} SO_3$.
↖ anhydride of H_2SO_4.

$HNO_3 \xrightarrow{-H_2O} N_2O_5$
↖ anhydride of HNO_3.

→ Basic oxides are anhydrides of metal hydroxides

Eg:- $2NaOH \xrightarrow{-H_2O} Na_2O$.
$Ba(OH)_2 \xrightarrow{-H_2O} BaO$.

Types of oxides:-

① Acidic oxides:- Generally oxides of non-metal which react with bases.
→ If soluble, form acidic sol^n (acidic sol^n turns blue litmus red).
Eg:- SO_3, SO_2, SiO_2, CO_2, N_2O_5, N_2O_3, Cl_2O_7 etc

② Basic oxides:- Generally oxides of metals which react with acids.
→ If soluble, form basic sol^n (basic sol^n turns red litmus blue.

Eg:- Na_2O, BaO, CaO, FeO, Fe_2O_3 etc.

③ Amphoteric oxides:- These oxides react with both acids & bases.

Eg:- BeO, Al_2O_3, ZnO, Ga_2O_3, V_2O_5, PbO, PbO_2, SnO, SnO_2, Ga_2O_5, As_2O_3, etc.

$Al_2O_3 \xrightarrow{HCl\ (aq)} AlCl_3(aq) + H_2O(l)$

$Al_2O_3 \xrightarrow{NaOH\ (aq)} NaAlO_2(aq)$ (Sod. metaluminate)

$V_2O_5 \xrightarrow{HCl\ (aq)} VO_2^+(aq) + Cl^-(aq) + H_2O$ (Oxycation)

$V_2O_5 \xrightarrow{NaOH\ (aq)} VO_4^{3-}(aq)$.

$ZnO \xrightarrow{HCl\ (aq)} ZnCl_2(aq) + H_2O(l)$

$ZnO \xrightarrow{NaOH\ (aq)} Na_2(ZnO_2)(aq) + H_2O$ (sod. zincate)

$Na[Zn(OH)_4]$

④ Neutral oxides:- Neutral towards acid as well as base, i.e. do not react with acids & bases.

Eg:- N_2O, NO, CO.

⑤ Mixed anhydrides:- Anhydride of 2 acids.

Eg:- $HNO_2 + HNO_3 \xrightarrow{-H_2O} N_2O_4$ or $2NO_2$.

ClO_2 is an anhydride of $HClO_2$ & $HClO_3$.

Trends of nature of oxides in periodic table →

→ Greater the metallic character, greater is the basic nature.

→ Along the period, basic nature of oxide decreases and acidic nature increases.

→ Down the group, basic nature of oxide increases and acidic nature decreases.

→ increasing acidic nature.

Eg:- $Na_2O < MgO < Al_2O_3 < SiO_2 < P_4O_{10} < SO_3 < Cl_2O_7$

basic — amphoteric — weakly acidic — strongly acidic

$Li_2O > BeO > B_2O_3$.

→ decreasing basic nature.

$Li_2O < Na_2O < K_2O < Rb_2O < Cs_2O$.

→ increasing basic nature.

$BeO < MgO < CaO < SrO < BaO$

→ increasing basic nature.

$Cl_2O > Br_2O > I_2O$

→ decreasing acidic nature.

$HClO > HBrO > HIO$.

→ decreasing acidic nature.

$HClO_3 > HBrO_3 > HIO_3$

→ decreasing acidic nature.

NOTE:- When an element forms oxy acids in two or more different oxidation states, then, in general greater the oxidation state, greater will be the acidic nature.

Eg:- $H_2SO_4 > H_2SO_3$ $\quad SO_3 > SO_2$

$HNO_3 > HNO_2$. $\quad NO_3 > NO_2$.

Eg:- $\overset{(+2)}{MnO} < \overset{(+3)}{Mn_2O_3} < \overset{(+4)}{MnO_2} < \overset{(+6)}{MnO_3} < \overset{(+7)}{Mn_2O_7}$

→ Increasing acidic nature.

Exceptions :-

$H_3\overset{(+3)}{P}O_3 \geq H_3\overset{(+1)}{P}O_2 > H_3\overset{(+5)}{P}O_4$ ⟶ decreasing acidic nature.

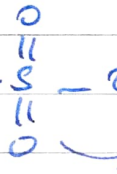
pK_a values : $1.1 \geq 1.2 > 2.1$.

→ Lower the pK_a values, greater the acidic strength.

Oxyacids :- where 'H' is attached to 'O' atom and 'O' is again attached to some other electronegative element.

Eg :- H_3PO_4, H_2SO_4, HNO_3, CH_3COOH, etc.

Eg :-



⟶ acidic hydrogen.

→ (dibasic or diprotic acid)

Hydracids (covalent hydrides) :- where 'H' is attached to some electronegative atom.

Eg :- HF, HCl, HBr, HI, H_2O, H_2S, H_2Se, NH_3, PH_3, etc.

→ Along the period, acidic nature of covalent hydride increases.

→ Down the period, acidic nature of covalent hydride increases.

Eg :- $CH_4 < NH_3 < H_2O < HF$, $SiH_4 < PH_3 < H_2S < HCl$

⟶ increasing acidic strength.

(In Pd., EN is determining factor.

EN↑, acidic nature ↑)

$HF < HCl < HBr < HI$

$H_2O < H_2S < H_2Se < H_2Te$ ⟶ increasing acidic strength.

$NH_3 < PH_3 < AsH_3 < SbH_3$.

(In a grp., acidic nature is determined by size of atom and bond dissociation energy.

Greater the size, lower will be the bond dissociation energy & greater will be the acidic nature.

Diagonal Relationship.

Li Be B C N
Na Mg Al Si P.

→ show many similar properties in their compounds & called diagonal relationship.

→ Reasons for diagonal relationship →

① same $\frac{\text{charge}}{\text{radius}}$ ratio of pair elements.

Eg:- (i) Both BeO & Al_2O_3 are amphoteric
(ii) Li_2CO_3 & $MgCO_3$ on thermal decomposition give CO_2 gas.
(iii) Both Be_2C & Al_4C_3 give CH_4 gas on hydrolysis